

ECON 100 - Introduction to Economics

Homework 5

- Define any variables you need to answer the problems.
- You may use any material from this course (including other students). You may use Gen AI to help check your work or understanding, but you should not copy-paste solutions or answers obtained from Chat GPT/AI.
- Define any variables you need to answer the problems.
- Keep your short-answer explanations short but clear. Your goal is to convince a skeptical grader that you understand the relevant concepts well enough to answer the question you are given.
- The questions on this homework sum to 70.
- Please be sure to write down the name(s) of anyone else you worked with on this homework (either at the top or under your name). Place a star next to your name OR underline OR some way to make it easy for me to tell which homework is whose.
- Bring any and all questions to office/student hours!

-
1. (2 points) What is your name?
 2. (2 points) What section are you in? (Either section number or what time your class meets)

1 True/False Questions

Indicate “T” rue or “F” alse for each of the following statements or claims. If you correctly explain why a statement or claim is false, I will give you a bonus point up to a 100% on this homework.

3. (2 points) Suzie is reviewing their costs and benefits from their lemonade stand in Clinton. They notice after graphing the data that between a quantity of 10 and 20 lemonades their marginal cost curve is increasing at the same time their average variable cost curve is decreasing. They conclude they must have done the math incorrectly for calculating marginal and/or average variable cost. **Claim:** Suzie’s logic is consistent with what we have learned in class regarding cost curves.
4. (2 points) Suppose that there are 1,000 hot pretzel stands operating in New York City. Each stand owns a license to operate. Each stand has a typical U-shaped ATC curve. The market demand curve for pretzels slopes downward and the market for pretzels is in long run competitive equilibrium. **Claim:** If the city decides to reduce the number of licenses to 800, the short and long-run profits of all 800 firms which remain will still be 0.
5. (2 points) **Claim:** If a competitive market is in long-run equilibrium, and market price is currently above a firm’s Average Variable Cost at the profit-maximizing quantity, we can say for certain the firm will remain in the market.
6. (2 points) **Claim:** According to our supply-demand model, there does not exist any form of binding governmental intervention that does not result in deadweight loss (i.e., any form of binding governmental intervention results in deadweight loss).
7. (2 points) Suppose a market is comprised of identical firms (with identical cost structures). In the long-run, the price in a perfectly competitive market will be equal to the minimum of the firm’s average total cost curve.

2 Multiple Choice Questions

Choose the best answer for each question. There is ONLY one correct answer for each question. No explanation necessary.

8. Suppose the market price for trimming hedges is \$100 per customer/yard, no matter the number of hedges someone needs trimmed. This is a perfectly competitive market. Hedge-Be-Gone, a company in New Hartford, has to pay \$50 per day, regardless of how many customers they serve in a given day. The company's variable costs for hedge trimming is as follows:

Customers/Yards	Variable Cost
1	200
2	260
3	320
4	360
5	420
6	500
7	600
8	720
9	860
10	1020

- (a) (2 points) Which of the following reflects Hedge-Be-Gone's profit maximizing quantity?
- A. 7.
 - B. 8.
 - C. 9.
 - D. 10.
- (b) (2 points) Assume Hedge-Be-Gone makes decisions consistent with profit-maximizing behavior as we have learned in class. At Hedge-Be-Gone's profit-maximizing quantity and market price, in the short-run Hedge-Be-Gone will _____ and in the long-run Hedge-Be-Gone will _____. Choose the option that correctly fills in the blanks.
- A. shutdown, remain.
 - B. shutdown, exit.
 - C. produce, remain.
 - D. produce, exit.
- (c) (2 points) Which of the following correctly predicts how the competitive market will adjust from the current equilibrium price to the long-run equilibrium price, assuming no other changes in the market for hedge trimming.
- A. Firms will exit, and equilibrium price will fall.
 - B. Firms will enter, and equilibrium price will fall.
 - C. Firms will exit, and equilibrium price will rise.
 - D. Firms will enter, and equilibrium price will rise.
9. (2 points) Bill's willingness to pay for chipotle is \$10 per bowl. When the price of a chipotle bowl is \$8, Bill purchases 5 bowls. Then chipotle raised the price of bowls to \$9 and Bill purchased 4 bowls after the price change. **What is the reduction in Bill's consumer surplus due to the price change?**
- A. \$2.
 - B. \$6.
 - C. \$0.
 - D. \$1.

10. (2 points) Suppose a perfectly competitive firm's average total cost curve is given by $ATC = q + \frac{100}{q}$, which means the slope of ATC is given by $1 - \frac{100}{q^2}$. Given this information, what is the perfectly competitive firm's long-run profit maximizing quantity?
- A. 1.
 - B. 10.
 - C. 100.
 - D. None of these.

3 Short Answer Questions

These questions all require math, diagrams, and/or written explanations. Remember you are trying to convince me you understand the why and how of what you are doing, not simply that you are able to get the numerical/graphical answer correct. Include specific concepts from class in your written explanations for full credit.

11. **Should Chick-fil-A Open in Clinton?** Suppose the market for a fried chicken sandwich in Clinton is perfectly competitive, with infinitely many options (LOL). Chick-Fil-A would be another perfectly competitive firm, and is trying to decide if they should open a location in Clinton or not, and you are tasked with helping them figure it out!
- (a) (4 points) **Market Research.** After some research, you estimate Clinton's market demand as $Q = 1200 - 50P$, and market supply as $Q = 50P$. Find equilibrium price and quantity in this market,
 - (b) (6 points) Draw the market supply-demand diagram for this market. Label your axes, and indicate the equilibrium price and quantity on your diagram.
 - (c) (4 points) Chick-Fil-A's total costs are given by $TC = 64 + q^2$ and $MC = 2q$. Next to your market supply-demand diagram, graph Chick-Fil-A's marginal cost, average total cost, and average variable cost functions.
 - (d) (6 points) What would Chick-Fil-A's marginal revenue be in this market? Explain briefly.
 - (e) (4 points) Calculate Chick-Fil-A's profit maximizing quantity, and Chick-Fil-A's profit at this quantity.
 - (f) (6 points) Should Chick-Fil-A open a Clinton location? Explain.
 - (g) (6 points) What is the lowest price for a chicken sandwich at which Chick-Fil-A would be willing to open a location in Clinton? Filling out the table below may be helpful.

q	Average Total Cost
2	
4	
8	
16	
32	

- (h) (10 points) **Completion Only.** Is this a realistic way of thinking about if Chick-Fil-A should enter or not? What assumptions about the market for chicken sandwiches might not be realistic? How might relaxing those assumptions change the answer?

4 Ungraded Questions

12. Suppose the book printing industry is competitive and begins in a long-run equilibrium.
- (a) (0 points) Draw a diagram showing a typical firm in the industry.
 - (b) (0 points) If Hi-Tech Printing Company invents a new process that sharply reduces the cost of printing books, what happens to Hi-Tech's profits and the price of books in the short run when Hi-Tech has a patent that prevents other firms from using the new technology?
 - (c) (0 points) What happens in the long run when the patent expires and other firms are free to use the technology?
13. Suppose that there are 1,000 hot pretzel stands operating in New York City. Each stand has a typical U-shaped ATC curve. The market demand curve for pretzels slopes downward and the market for pretzels is in long run competitive equilibrium.
- (a) (0 points) Draw the current equilibrium, using graphs for the entire market and the an individual pretzel stand.
 - (b) (0 points) The city decides to restrict pretzel stand licenses, reducing the number fo only 800. What effect will this have on the market and on an individual stand that is still operating? Draw graphs to illustrate your answer.
 - (c) (0 points) Suppose that the city decides to charge a fee for the 800 licenses, all of which are quickly sold. How will the size of the fee affect the number of pretzels sold by an individual stand? How will it affect the price of pretzels in the city?
 - (d) (0 points) The city wants to raise as much revenue as possible, while ensuring that all 800 licenses are sold. How high should the city set the license fee? Show your answer on your graph.
14. (0 points) Bob's lawn-mowing service is a profit-maximizing competitive firm. Bob mows lawns for \$27 each. His total cost each day is \$280, of which \$30 is a fixed cost. He mows 10 lawns per day. What can you say about his short-run decision regarding shut down and his long run decision regarding exit?